

● Nature of science: Theories

Law or principle?

Many physics books refer to the 'law' of conservation of energy, others refer to the 'principle' of conservation of energy. Is there any difference? And why is it not called the 'theory' of conservation of energy?

Research to determine if there are any differences between 'theories', 'laws' and 'principles.'

- 5 Outline the energy transfers that occur when a mass hanging vertically on the end of a metal spring is displaced and allowed to move up and down (oscillate) freely until it stops moving.
- 6 State the main energy transfers that occur in the use of a mobile phone.
- 7 State the names of devices whose main uses are to perform the following energy transfers:
 - a electricity to sound
 - b chemical to electricity
 - c sound to electricity
 - d chemical to radiation
 - e chemical to kinetic
 - f elastic to kinetic
 - g kinetic to electricity
 - h chemical to internal
 - i electromagnetic radiation to electricity.
- 8 Sketch and annotate a Sankey diagram to represent the following process. A car transfers 1500 kJ from its fuel as it gains 100 kJ of kinetic energy and 200 kJ of gravitational potential energy. The rest of the energy is dissipated into the environment.
- 9
 - a What are the two main types of energy of an aircraft flying at a height of 12 km?
 - b After the airplane has landed at an airport, what has happened to most of that energy?
- 10 An adult male body transfers about 10^7 J of energy every day.
 - a Name the source of this energy.
 - b
 - i Outline the principle uses for this energy and
 - ii why does it have to be replaced?
- 11 A car slows down for traffic lights. State two causes of energy dissipation.

● Common mistake

Many students believe (wrongly) that batteries store electrical energy. Batteries contain chemical compounds that react when a circuit containing the battery is connected. Chemical energy is then transferred to an electric current. Many batteries can be recharged if the direction of current is reversed. This reverses the chemical changes.

Calculating mechanical energies

SYLLABUS CONTENT

- ▶ Mechanical energy is the sum of kinetic energy, gravitational potential energy and elastic potential energy.
- ▶ Work done by the resultant force on a system is equal to the change in the energy of the system.
- ▶ Kinetic energy of translational motion as given by: $E_k = \frac{1}{2}mv^2 = \frac{p^2}{2m}$.
- ▶ Gravitational potential energy, when close to the surface of the Earth as given by: $\Delta E_p = mg\Delta h$.
- ▶ Elastic potential energy as given by: $E_{\text{H}} = \frac{1}{2}k\Delta x^2$.

◆ **Mechanical energy**
Energy of a macroscopic object which can do useful work.

Work can be done on a macroscopic object / system to give it kinetic energy, gravitational potential energy, or elastic potential energy. The object then has the ability to transfer that energy to do useful work. We say that it has **mechanical energy**. We will now show how these three types of energy can be calculated.

The symbol E can be used to represent energy, usually with a subscript to represent the particular type of energy. (Note: E is also used for electric field.)

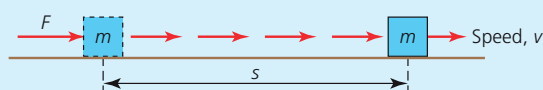
Kinetic energy

Tool 3: Mathematics

Derive relationships algebraically

Kinetic energy can be calculated from the equation $E_k = \frac{1}{2}mv^2$. This is explained / derived below. (Ideally, we would like to fully explain the origin of all of the equations used in this course, but that is not always possible.)

Consider a mass m accelerated horizontally from rest by a constant resultant force, F , acting in the direction of motion, as shown in Figure A3.16.



■ **Figure A3.16** Doing work to increase movement

Using the equation of motion $v^2 = u^2 + 2as$ (from Topic A.1) and noting that, in this example, $u = 0$, we see that the distance travelled, s , can be determined from the equation: $s = \frac{v^2}{2a}$.

The work done W , in a distance s , can be calculated from: $W = Fs = ma \times \frac{v^2}{2a} = \frac{1}{2}mv^2$.

This amount of energy has been transferred from the origin of the force to the moving mass. We say that the mass has *gained* kinetic energy, E_k .

$\frac{1}{2}mv^2$ can also be written as: $\frac{[mv]^2}{2m}$.

Since momentum, mv , is given the symbol p , kinetic energy can also be determined from the equation: $E_k = \frac{p^2}{2m}$.

This version is most commonly used with the kinetic energy and momentum of atomic particles (See Topic E.5).

The kinetic energy of a moving mass can be calculated using: $E_k = \frac{1}{2}mv^2$

Or from: $E_k = \frac{p^2}{2m}$



◆ **Order of magnitude** An approximate value rounded to the nearest power of ten.

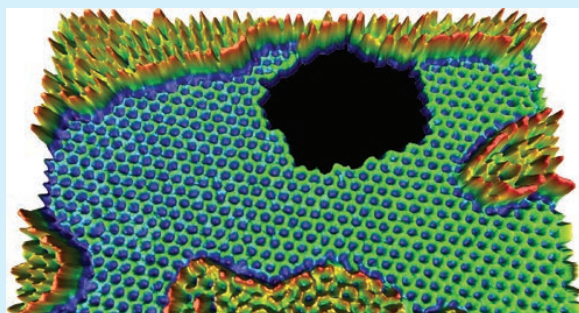
Some typical examples of kinetic energies (to an **order of magnitude**):

- a molecule in air at 20° C: 10^{-20} J
- a falling snowflake: 10^{-7} J
- a boy running in a 100 m race: 10^3 J
- a bullet from a rifle: 10^4 J
- a car moving along an open road: 10^5 J
- a large meteor moving towards Earth: 10^{17} J
- the Earth in its motion around the Sun: 10^{33} J.

Tool 3: Mathematics

Compare and quote values and approximations to the nearest order of magnitude

Physics is the fundamental science that tries to explain how and why everything in the Universe behaves in the way that it does. Physicists study everything from the smallest parts of atoms to distant objects in our galaxy and beyond (Figure A3.17).



■ **Figure A3.17** **a** The arrangement of individual atoms in graphene (a material made from a single layer of carbon atoms) can be seen using a special type of electron microscope; **b** complex gas and dust clouds in the Cat's Eye nebula, 3000 light-years away

Measurements and calculations commonly relate to the world that we can see around us (the macroscopic world), but our observations may require microscopic explanations, often including an understanding of molecules, atoms, ions and subatomic particles. Astronomy is a branch of physics that deals with the other extreme – quantities that are very much bigger than anything we experience in everyday life.

The study of physics therefore involves dealing with both very large and very small numbers. When numbers are so different from our everyday experiences, it can be difficult to appreciate their true size. For example, the age of the Universe is believed to be about 10^{18} s, but just how big is that number?

When comparing quantities of very different sizes (magnitudes), for simplicity we often make approximations to the nearest power of 10. When numbers are approximated and quoted to the nearest power of 10, it is called giving them an order of magnitude. For example, when comparing the lifetime of a human (the worldwide average is about 70 years) with the age of the Universe (1.4×10^{10} y), we can use the approximate ratio $10^0 / 10^2$. That is, the age of the Universe is about 10^8 human lifetimes, or we could say that there are eight orders of magnitude between them.

Here are three further examples:

- The mass of a hydrogen atom is 1.67×10^{-27} kg. To an order of magnitude this is 10^{-27} kg.
- The distance to the nearest star (*Proxima Centauri*) is 4.01×10^{16} m. To an order of magnitude this is 10^{17} m. (Note: $\log$ of $4.01 \times 10^{16} = 16.60$, which is nearer to 17 than to 16.)
- There are 86 400 seconds in a day. To an order of magnitude this is 10^5 s.

Tables A3.1, A3.2 and A3.3 list the ranges of mass, distance and time that occur in the Universe.

■ **Table A3.1** The range of masses in the Universe

Object mass / kg	Object mass / kg
the observable Universe	10^{53}
our galaxy (the Milky Way)	10^{42}
the Sun	10^{30}
the Earth	10^{24}
a large passenger airplane	10^5
a large adult human	10^2
a large book	1
a raindrop	10^{-6}
a virus	10^{-20}
a hydrogen atom	10^{-27}
an electron	10^{-30}

■ **Table A3.2** The range of distances in the Universe

Distance size / m	Distance size / m
distance to the edge of the visible Universe	10^{27}
diameter of our galaxy (the Milky Way)	10^{21}
distance to the nearest star	10^{16}
distance to the Sun	10^{11}
distance to the Moon	10^8
radius of the Earth	10^7
altitude of a cruising airplane	10^4
height of a child	1
how much human hair grows by in one day	10^{-4}
diameter of an atom	10^{-10}
diameter of a nucleus	10^{-15}

■ **Table A3.3** The range of times in the Universe

Time period time interval / s	Time period time interval / s
age of the Universe	10^{18}
time since dinosaurs became extinct	10^{15}
time since humans first appeared on Earth	10^{13}
time since the pyramids were built in Egypt	10^{11}
typical human lifetime	10^9
one day	10^5
time between human heartbeats	1
time period of high-frequency sound	10^{-4}
time for light to travel across a room	10^{-8}
time period of oscillation of a light wave	10^{-15}
time for light to travel across a nucleus	10^{-23}

Estimation

Sometimes we do not have the data needed for accurate calculations, or maybe calculations need to be made quickly. Sometimes a question is so vague that an accurate answer is simply not possible. The ability to make sensible estimates is a very useful skill that needs plenty of practice.

When making estimates, different people will produce different answers and it is usually sensible to use only 1 (maybe 2) significant figures. Sometimes only an order of magnitude is needed.

The numbers in the list of kinetic energies given above cannot be given with precision because the situations are vague and there are a wide range of possibilities (with the exception of the Earth's kinetic energy). For example, boys (of all ages) in 100 m races could have masses between 20 kg and 70 kg (or more), and they could run at speeds between 2 m s^{-1} and 9 m s^{-1} . These figures correspond to a kinetic energy range of 90–2800 J. To give a value with 2, or 3, significant figures would be misleading. It is more sensible to give a typical value to the nearest power of 10 (or sometimes, 1 significant figure). In the case of boys running 100 m, values of 10^2 J or 10^3 J may be considered typical (a matter of opinion).

WORKED EXAMPLE A3.4

A constant resultant horizontal force of 40 N accelerated a box over a distance of 50 cm.

- How much work was done on the box?
- State the assumption that you made answering part a.
- Calculate the kinetic energy that was gained by the box.
- If the box was initially at rest and then reached a speed of 2.9 m s^{-1} , what was its mass?

Answer

- $W = Fs = 40 \times 0.50 = 20 \text{ J}$
- The force was in the same direction as the motion of the box.
- 20 J
- $$E_k = \frac{1}{2}mv^2$$

$$20 = \frac{1}{2} \times m \times 2.9^2$$

$$m = \frac{(2 \times 20)}{2.9^2}$$

$$= 4.8 \text{ kg}$$

Tool 3: Mathematics

Understand the significance of uncertainties in raw and processed data

Although scientists are perceived as working towards finding ‘exact’ answers, an unavoidable **uncertainty** exists in every measurement. The results of all scientific investigations have uncertainties and errors, although good experimenters will try to keep these as small as possible.

When we receive numerical data of any kind (scientific or otherwise) we need to know how much belief we should place in the information that we are reading or hearing. The presentation of the results of serious scientific research should always have an assessment of the uncertainties in the findings, because this is an integral part of the scientific process. Unfortunately, the same is not true of much of the information we receive through the media, where data is too often presented uncritically and unscientifically, without any reference to its source, uncertainties or reliability.

No matter how hard we try, even with the very best of measuring instruments, it is simply not possible to measure anything exactly. For one reason, the things that we can measure do not exist as perfectly exact quantities; there is no reason why they should.

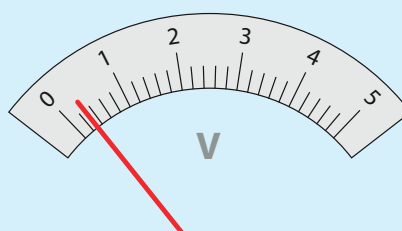
This means that every measurement is an approximation. A measurement could be the most accurate ever made, for example the width of a ruler might be stated as 2.283 891 03 cm, but that is still not perfect, and even if it was, we would not know because we would always need a more accurate instrument to check it. In this example we also have the added complication of the fact that when measurements of length become very small, we have to deal with the atomic nature of the objects that we are measuring.

A measurement is **accurate** if it is close to its true value. For example, if you weigh a mass of 90.1 g and the result is 90.1 g, then the measurement can be described as accurate. However, be aware that in scientific research, true values are usually not known.

If the same mass (90.1 g) was measured by a different method, or a different person and the result was 89.8 g, then there was a clear **error** in the measurement. An error has occurred if the measurement is different from its true value (using an appropriate number of significant figures for the comparison).

Significant errors are often due to faulty apparatus or poor experimental skills. It is often possible to correct the source of such errors.

A **systematic error** occurs because there is something consistently wrong with the measuring instrument or the method used. A reading with a systematic error is always either bigger or smaller than the correct value by the same amount. Common causes are instruments that have an incorrect scale (wrongly calibrated), or instruments that have an incorrect value to begin with, such as a meter that displays a reading when it should read zero. This is called a **zero-offset error** – an example is shown in Figure A3.18. A thermometer that incorrectly records room temperature will produce systematic errors when used to measure other temperatures.



■ **Figure A3.18** This voltmeter has a zero-offset error of 0.3 V, so that all readings will be too large by this amount.

Uncertainties in measurements are an indication of the amount of variation seen in the readings taken (without considering their accuracy). For example, if you weigh the same (unknown) mass five times you might get the following results: 53.2 g, 53.4 g, 52.9 g, 53.0 g, and 53.1 g. The results are not all the same, so there is clearly some random uncertainty in the results. The uncertainty in the use of the measuring instrument itself is usually assumed to be equal to the smallest division on its scale / display. In this example this is ± 0.1 g.

All experimental data has uncertainties. Sometimes uncertainties will arise because of difficulties in taking measurements. For example, human reaction times affect measurements made when using a stopwatch, or measuring the distance moved by something moving quickly can be difficult.

It is usually not possible to reduce uncertainties using the same apparatus and techniques.

In science the word **precise** means that measurements have low uncertainty. If, and when, measurements are repeated they will produce similar results. Consider Figure A3.19, which shows where arrows fired at a target (not shown) landed on eight separate occasions. Because we do not know where the target is, we cannot tell if the arrows were fired accurately, or not. But we can describe **b** and **d** as more precise than **a** and **c**.

◆ **Uncertainty (random)**

The range, above and below a stated value, over which we would expect any repeated measurements to occur. Uncertainty can be expressed in absolute, fractional or percentage terms.

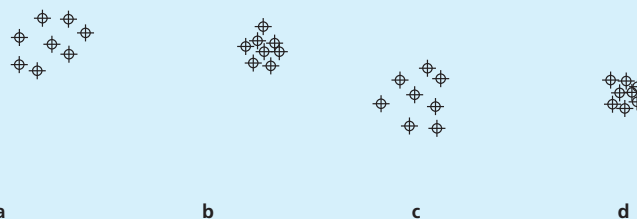
◆ **Accuracy** A single measurement is described as accurate if it is close to the correct result. A series of measurements of the same quantity can be described as accurate if their mean is close to the correct result.

◆ **Error** When a measurement is not the same as the correct value.

◆ **Systematic error** An error which is always either bigger or smaller than the correct value by the same amount, for example a zero-offset error.

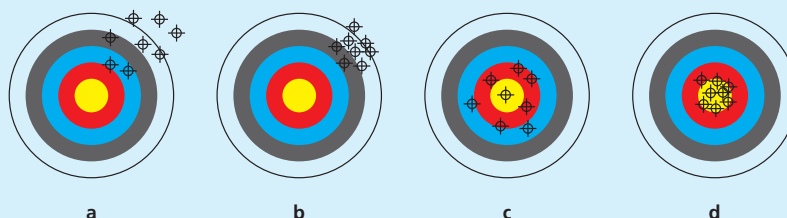
◆ **Zero offset error** A measuring instrument has a zero offset error if it records a non-zero reading when it should be zero.

◆ **Precision** A measurement is described as precise if a similar result would be obtained if the measurement was repeated.



■ **Figure A3.19** Precision of arrows hitting unseen target

In Figure A3.20 the positions of the targets have been included and we can see that the accuracy was good in **d**. But note that the accuracy seen in **c** is also good, because the average position of the arrows is close to the centre of the target.



■ **Figure A3.20** Accuracy of arrows hitting target



Common mistake

Uncertainties and errors are often confused, and different sources may define them slightly differently. See **Tool 3: Mathematics (Propagating uncertainties)** on page 131.

WORKED EXAMPLE A3.5

A student wanted to determine the increase of a trolley's kinetic energy as it accelerated down a slope. The trolley had a mass of $576 \text{ g} \pm 5 \text{ g}$ and its length was $28.0 \text{ cm} \pm 0.5 \text{ cm}$. Using a stopwatch the student measured the time for the trolley to pass a point near the top of the slope to be 1.26 s . Near the bottom of the slope the trolley took 0.73 s to pass a particular point. Because it was difficult to start and stop the stopwatch at exactly the right time, it was estimated that the uncertainty in each time measurement was 0.10 s .

Calculate a value for the increase in kinetic energy of the trolley. Determine the absolute and percentage uncertainties in the answer. Comment on your answer

Answer

Near the top:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.576 \times \left(\frac{0.28}{1.26}\right)^2 = 1.42 \times 10^{-2} \text{ J}$$

- Percentage uncertainty in mass = $100 \times (5/576) = 0.87\%$
- Percentage uncertainty in length = $100 \times (0.5/28) = 1.79\%$
- Percentage uncertainty in time = $100 \times (0.10/1.26) = 7.94\%$
- Percentage uncertainty in kinetic energy = $0.87 + 1.79 + 1.79 + 7.94 + 7.94 = 20.33\%$
- Absolute uncertainty in kinetic energy = $(20.33/100) \times (1.42 \times 10^{-2}) = 0.29 \times 10^{-2} \text{ J}$

Near the bottom:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.576 \times \left(\frac{0.28}{0.73}\right)^2 = 4.24 \times 10^{-2} \text{ J}$$

- Percentage uncertainty in mass = $100 \times (5/576) = 0.87\%$
- Percentage uncertainty in length = $100 \times (0.5/28) = 1.79\%$
- Percentage uncertainty in time = $100 \times (0.10/0.73) = 13.70\%$
- Percentage uncertainty in kinetic energy = $0.87 + 1.79 + 1.79 + 13.70 + 13.70 = 31.85\%$
- Absolute uncertainty in kinetic energy = $(31.85/100) \times 4.24 \times 10^{-2} = 1.35 \times 10^{-2} \text{ J}$
- Difference in kinetic energies = $(4.24 \times 10^{-2}) - (1.42 \times 10^{-2}) = 2.82 \times 10^{-2} \text{ J}$
- Absolute uncertainty in difference = $(0.29 \times 10^{-2}) + (1.35 \times 10^{-2}) = 1.64 \times 10^{-2} \text{ J}$
- Percentage uncertainty in difference = $\left(\frac{1.64}{2.82}\right) \times 100 = 58\%$

This high percentage uncertainty in the results of this experiment may be surprising. The experiment needs redesigning if accurate results are needed.

Other types of kinetic energy

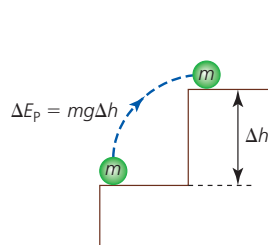
We can only use $E_k = \frac{1}{2}mv^2$ to determine the translational kinetic energy of objects travelling from place to place. Objects which are vibrating or rotating also have kinetic energy, but we need different equations to calculate their values. This will be covered in Topics A.4 and C.1 (for HL students).

- 12** Calculate the kinetic energy of a 57 g tennis ball served with a speed of 50 ms^{-1} .
- 13 a** Determine the work needed to be done on a 1800 kg car to accelerate it from rest to 20 ms^{-1} .
- b** What average resultant force is needed to do this in a horizontal distance of 100 m?
- c** If the car decelerates from the same speed to rest in 70 m, calculate the average force exerted on the car
- d** What are the locations of this force?
- 14 a** Calculate the kinetic energy of an electron (mass = $9.110 \times 10^{-31} \text{ kg}$) moving with a speed which is 5% of the speed of light ($3.0 \times 10^8 \text{ ms}^{-1}$).
- b** What is the momentum of this electron?
- 15** Suppose that the force on the box in Worked example A3.4 continued to act for another 50 cm. Determine the final speed of the box after it had moved a total distance of 1.0 m.

- 16 a** Calculate the kinetic energy of a high-speed train (see Figure A3.21) which has a mass of $7.5 \times 10^5 \text{ kg}$ and is moving with a speed of 300 km h^{-1} .
- b** Compare your answer to the kinetic energy of a typical family car on a motorway.
- c** What average force is needed to reduce the speed of the train uniformly to zero in a distance of 5.0 km?
- d** Suggest two reasons why high-speed railway systems do not have many stations.



■ **Figure A3.21** A high-speed train



■ **Figure A3.22** A mass gaining gravitational potential energy



Gravitational potential energy

Consider raising a mass m a vertical height Δh , as shown in Figure A3.22. The minimum force required is equal to the weight of the mass, $F_g = mg$.

The work done by the force on the mass, $W = Fs = mg\Delta h$

This amount of energy has been transferred from the origin of the force to the raised mass. We say that the mass has gained gravitational potential energy, ΔE_p . If the mass is allowed to fall back down the same distance, the same amount of energy could be transferred to do useful work. Falling water in a **hydroelectric power** station uses this principle.

The gravitational potential energy of a mass raised a height Δh , close to the Earth's surface can be calculated using $\Delta E_p = mg\Delta h$

◆ Hydroelectric power

The generation of electrical power from falling water.

LINKING QUESTION

- Why is the equation for the change in gravitational potential energy only relevant close to the surface of the Earth, and what happens when moving further away from the surface?

This is discussed in Topic D.1 for HL students.

● Top tip!

A mass resting on a table, or on the ground, does not have *zero* gravitational potential energy. When we use $\Delta E_p = mg\Delta h$ we are calculating how much more, or less, gravitational potential energy the object has in its new position compared to the place from where it was moved.

In other words, we are calculating a *change* in gravitational potential energy.

In the detailed study of gravitational fields (Topic D.1) HL students will need to consider if there is any place where an object really does have zero gravitational energy.

WORKED EXAMPLE A3.6

Estimate the gravitational potential energy gained by a teenage girl who moves from ground level to the viewing platform on the 124th floor of the Burj Khalifa in Dubai (see Figure A3.23).

Assume that the girl has a mass of 40 kg and the height of each floor is 3.5 m.



■ **Figure A3.23** The Burj Khalifa in Dubai

Answer

$$\Delta E_p = mg\Delta h = 40 \times 9.8 \times (124 \times 3.5) \approx 2 \times 10^5 \text{ J}$$

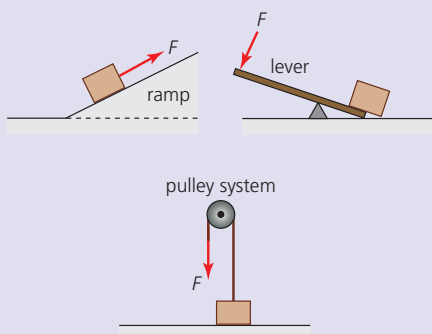
ATL A3A: Research skills

Use search engines and libraries effectively; evaluate information sources for accuracy, bias, credibility and relevance

When lifting a heavy object, the amount of gravitational potential energy that we need to transfer to it is decided only by its weight and the vertical height ($\Delta E_p = mg\Delta h$). For example, when a 50 kg mass is raised a height of 1 m it gains about 500 J of gravitational potential energy. Although 500 J may not be a lot of work to do, that does not mean that we can do this job easily.

There are two main reasons why this job could be difficult. Firstly, we may not be able to transfer that amount of energy in the time required to do the work. Another way of saying this is that we may not be powerful enough. (Power is discussed later in this topic.) Secondly, we may not be strong enough because we are not able to provide the required upwards force of 500 N. Power and strength are often confused with each other in everyday language.

Lifting (heavy) weights is a common human activity and many types of simple ‘machine’ were invented many years ago to make this type of work easier, by reducing the force needed. These include the ramp (inclined plane), the **lever** and the **pulley** (Figure A3.24).



■ **Figure A3.24** Simple machines which can be used to raise loads

◆ **Lever** A simple machine consisting of a rigid bar and a pivot. Used to change the direction and magnitude of a force.

◆ **Pulley** A rotating wheel used to change the direction of a force. When two or more pulleys are combined, the system can reduce the force needed to do work.

Elastic potential energy

When a spring, or an elastic material, is deformed from the shape it had when there was no force acting on it, as in Figure A3.26, it will become a store of elastic potential energy that can later be used to transfer useful energy when the spring / material returns to its original shape.

Remember Hooke’s law for elastic stretching from Topic A.2: $F_H = -kx$, where k is the ‘spring constant’. Note also that, if a force on a spring / elastic material increases from 0 to F_H , the average force used during the deformation is $\frac{1}{2}F_H$, assuming that the deformation is proportional to the force (as shown in Figure A3.26).

In each of these simple machines the force needed to do the job is reduced, but the distance moved by the force is increased. If there was no energy dissipation (mainly due to friction), the work done by the force (Fs) would equal the useful energy transferred to the object being raised ($mg\Delta h$). In practice, because of energy dissipation, we will transfer more energy using a machine than if we lifted the load directly, without the machine. However, this is not a problem because we are usually much more concerned about how easy it is to do a job, rather than the total energy needed, or the efficiency of the process.

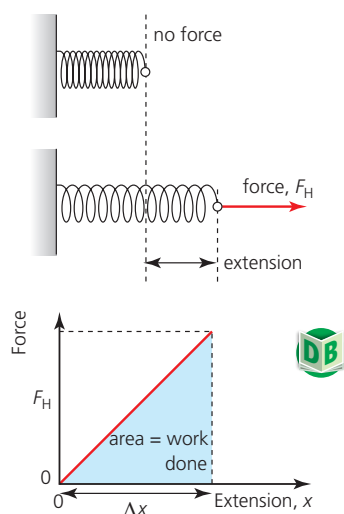
Figure A3.25 shows another example of a simple machine: a car jack being used to raise one side of a car.



■ **Figure A3.25** Changing a car tyre using a simple machine (car jack)

By changing the design of a car jack, it is possible in theory to raise the car with any sized force that we choose. For example, a force of 50 N may raise the wheel off the ground if the handle is rotated 10 times, whereas a force of 10 N would require about 50 rotations of the handle to transfer the same amount of energy. The heavy weight of the car will produce a lot of friction in the car jack.

The construction of many ancient structures was only possible because of the simple machines seen in Figure A3.24. Research online to learn some details of how the pyramids (or any other building of a similar age) were made. Compare the information you retrieve from different sources. Do the sources agree? Which sources do you consider most credible? Why?



■ **Figure A3.26** Force and extension when stretching a spring

For a spring / material which obeys Hooke's law, the work done, W , when it is deformed a distance Δx by a force F_H , is given by:

$$W = \text{average force} \times \text{distance} = \frac{1}{2}F_H \times \Delta x = \frac{1}{2}(k\Delta x)x = \frac{1}{2}k\Delta x^2$$

The work done is equal to the stored elastic potential energy, E_H . (There is no need to include a negative sign because work is not a vector quantity.)

The elastic potential energy of a deformed spring / material which obeys Hooke's law can be calculated by using:

$$E_H = \frac{1}{2}k\Delta x^2.$$

As explained earlier in this topic, the work done is equal to the area under force–extension graph.

WORKED EXAMPLE A3.7

An open-coiled compression spring, like the one seen in Figure A3.27 has an overall length of 5.64 cm. A compressive force of 59.0 N reduced its length to 5.30 cm.

- Determine the spring constant, assuming that Hooke's law was obeyed.
- Calculate how much work was done to compress the spring.
- Use $E_H = \frac{1}{2}k\Delta x^2$ to confirm the amount of elastic potential energy stored in the spring.

Answer

- $k = \frac{F_H}{x} = \frac{59.0}{(5.64 - 5.30)} = 174 \text{ N cm}^{-1} \text{ (or } 1.74 \times 10^4 \text{ N m}^{-1})$
- $W = \frac{1}{2}F_H \Delta x = 29.5 \times (5.64 - 5.30) \times 10^{-2} = 0.100 \text{ J}$
- $E_H = \frac{1}{2}k\Delta x^2 = 0.5 \times (1.74 \times 10^4) \times (3.4 \times 10^{-3})^2 = 0.100 \text{ J}$

The answers to **b** and **c** are the same, as we would expect.

See Figure A3.6 for an example of how to determine elastic potential energy from a force–extension graph.



■ **Figure A3.27** An open-coiled compression spring

17 How much potential energy is transferred when:

- a** a 1.2 kg box is raised from the floor to a table top 0.85 m higher
- b** a 670 g book falls to the floor from the same table?

18 A cable car rises a vertical height of 700 m in a total distance travelled of 6.0 km.

- a** Show that approximately 15 MJ of gravitational potential energy must be transferred to a car of mass 1800 kg during the journey if it has six passengers with an average mass of 47 kg.
- b** Suggest why considerably more energy (than your answer to part **a**) has to be transferred in making this journey.



■ **Figure A3.28** Ngong Ping cable car in Hong Kong

19 A rocket launches a 500 kg satellite to a height of 400 km above the Earth's surface.

- a** Outline why the equation $\Delta E_p = mg\Delta h$, with $g = 9.8 \text{ N kg}^{-1}$ cannot be used to accurately determine the gravitational potential energy that has to be transferred to the satellite.
- b** However, the actual value of g at that height has not reduced as much as many students expect: during the launch it reduced from 9.8 N kg^{-1} to a value of 8.7 N kg^{-1} at a height of 400 km. Predict the gravitational potential energy gained by the satellite.

20 A spring has a spring constant of 384 N m^{-1} and is stretched by 2.0 cm.

- a** Calculate the elastic potential energy stored in the spring. Assume that it obeys Hooke's law.
 - b** Predict a value for the extension that would be needed to store 1.0 J of energy.
 - c** Explain why the answer to part **b** is uncertain.
- 21 a** Calculate the work done in raising the centre of gravity of a trampolinist of mass 62 kg through a vertical height of 3.48 m (see Figure A3.29).
- b** When he lands on the trampoline he is brought to rest for a moment before being pushed up in the air again. If the maximum displacement of the trampoline is 0.90 m, sketch a possible force–displacement graph for the surface of the trampoline.



■ **Figure A3.29** The more the trampoline stretches, the higher the trampolinist can jump